Journal Supplementary D — Full mathematical derivations

Companion to: *Beyond Accuracy* (Section IV)

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A.1 Scope

This supplement provides the complete mathematical content for the twelve-axis framework: generative models, variational inference, hierarchical Bayesian likelihood, coherence formulae, ARI and Hungarian definitions, and the algorithmic pseudocode for each axis.

B.2 Canonical generative model

B.2.1 LDA on band-frequency documents

For K topics, vocabulary $V = B$ band indices, D documents and hyperparameters $\alpha, \eta > 0$:

$$\phi_k \sim \mathrm{Dir}(\eta \mathbf{1}_V), \quad k = 1, \ldots, K, \tag{1}$$
$$\theta_d \sim \mathrm{Dir}(\alpha \mathbf{1}_K), \quad d = 1, \ldots, D, \tag{2}$$
$$z_{d,n} \sim \mathrm{Cat}(\theta_d), \quad n = 1, \ldots, N_d, \tag{3}$$
$$w_{d,n} \sim \mathrm{Cat}(\phi_{z_{d,n}}). \tag{4}$$

B.2.2 Online VB lower bound

The mean-field variational family is $q(\theta_d, z_d, \phi) = q(\theta_d \mid \gamma_d)\big(\prod_n q(z_{d,n} \mid \lambda_{d,n})\big) \prod_k q(\phi_k \mid \beta_k)$. The per-document ELBO is

$$\begin{aligned}
\mathcal{L}_d = \sum_n \sum_k \lambda_{d,n,k} \big[&\psi(\gamma_{d,k}) - \psi(\textstyle\sum_{k'} \gamma_{d,k'}) \\
&+ \psi(\beta_{k,w_{d,n}}) - \psi(\textstyle\sum_v \beta_{k,v}) - \log \lambda_{d,n,k} \big] \\
&+ (\alpha - 1) \sum_k \big[\psi(\gamma_{d,k}) - \psi(\textstyle\sum_{k'} \gamma_{d,k'}) \big] \\
&+ \log \Gamma(\textstyle\sum_{k'} \gamma_{d,k'}) - \textstyle\sum_k \log \Gamma(\gamma_{d,k}) + \text{const.}
\end{aligned} \tag{5}$$

Updates are Eq. (8) and Eq. (9) of the conference supplement B. Online VB applies the natural-gradient ascent $\beta \leftarrow (1 - \rho_t)\beta + \rho_t \tilde{\beta}_t$ with $\rho_t = (\tau_0 + t)^{-\kappa}$, $\tau_0 = 10$, $\kappa = 0.7$.

B.2.3 Posterior summaries

$\phi_{k,b} = \beta_{k,b} / \sum_v \beta_{k,v}$; $\theta_{d,k} = \gamma_{d,k} / \sum_{k'} \gamma_{d,k'}$; $z_d = \arg\max_k \theta_{d,k}$.

C.3 Axis F-1: hierarchical Bayesian model

C.3.1 Likelihood

For M methods, S scenes, F folds, observed fold-OA scores $y_{m,s,f}$:

$$
\begin{aligned}
y_{m,s,f} &= \mu_m + \delta_s + \rho_f + \epsilon_{m,s,f}, &(6)\\
\mu_m &\sim \mathcal{N}(0,1), &(7)\\
\delta_s &\sim \mathcal{N}(0,0.5^2), &(8)\\
\rho_f &\sim \mathcal{N}(0,0.2^2), &(9)\\
\epsilon_{m,s,f} &\sim \mathcal{N}(0,\sigma^2), &(10)\\
\sigma &\sim \mathrm{HalfNormal}(0.5). &(11)
\end{aligned}
$$

C.3.2 Inference

NUTS with 1000 draws + 1000 tune, 2 chains, $\hat{R} < 1.01$ on all parameters in all reported runs. Posterior mean and 94% HDI of μ_m are reported; pairwise $P[\mu_A > \mu_B]$ are estimated from the joint posterior draws as

$$
P[\mu_A > \mu_B] \approx \frac{1}{T}\sum_{t=1}^{T} \mathbf{1}[\mu_A^{(t)} > \mu_B^{(t)}], \tag{12}
$$

with $T = 2000$ draws across two chains.

D.4 Axis F-2: coherence measures

D.4.1 c_v coherence

For a topic t with top-N words $W_t = (w_1, \ldots, w_N)$, the NPMI between words w_i, w_j on a sliding window of size s is

$$
\mathrm{NPMI}(w_i, w_j) = \frac{\log \frac{P(w_i,w_j)+\epsilon}{P(w_i)P(w_j)}}{-\log(P(w_i,w_j)+\epsilon)}. \tag{13}
$$

The topic context vector is $\mathbf{v}_t = \sum_{i=1}^{N} \mathbf{u}_i$ with $u_{i,j} = \mathrm{NPMI}(w_i, w_j)$. The c_v score is

$$
c_v(t) = \frac{1}{N}\sum_{i=1}^{N} \cos(\mathbf{v}_t, \mathbf{u}_i), \tag{14}
$$

the average cosine similarity between the topic context vector and each word's individual context vector. We use $N = 15$ and the default Röder sliding window of size $s = 110$.

D.4.2 NPMI and U-Mass

$c_{\mathrm{NPMI}}(t) = \binom{N}{2}^{-1} \sum_{i<j} \mathrm{NPMI}(w_i, w_j)$. $\mathrm{U-Mass}(t) = \binom{N}{2}^{-1} \sum_{i<j} \log \frac{N(w_i,w_j)+1}{N(w_j)}$.

E.5 Axis F-3: seed stability

For each (scene, method): refit five LDA / ProdLDA / ETM under seeds $\{0, \ldots, 4\}$. For each refit compute KMeans clustering of θ into K clusters; report ARI(z_{KMeans}, label), c_v. Aggregate across seeds as mean ± std.

F.6 Axis F-4: capacity sensitivity

Refit LDA at $K \in \{4, 6, 8, 10, 12, 16, 20, 24\}$ with the canonical hyperparameters, $N = 5$ seeds per K. Per-K aggregates:

- $\text{perplexity}_K^{\text{test}} = \exp(-\frac{1}{|\mathcal{D}_{\text{test}}|}\sum_{d\in\mathcal{D}_{\text{test}}} \log p(\mathbf{w}_d \mid \phi_K, \alpha))$,
- $\text{topic-diversity}_K = \frac{|\bigcup_k W_k^{(K)}|}{N \cdot K}$ with $W_k^{(K)}$ the top-N words per topic at capacity K,
- matched-cosine_K = mean over (seed pair, topic) of $\cos(\phi_k^{(s_1)}, \phi_{\sigma^*(k)}^{(s_2)})$.

F.6.1 Canonical-K rule vs F-4 recommended K

The canonical fit used everywhere outside F-4 fixes

$$K_{\text{canonical}} = \max\big(4,\ \min(12,\ n_{\text{classes}})\big), \tag{15}$$

yielding $K = 12$ on Indian Pines, Salinas, KSC and Botswana, $K = 9$ on Pavia U, and $K = 6$ on Salinas-A. The F-4 capacity sweep, by contrast, returns $K_{\text{recommended}} = 4$ uniformly across all six labelled scenes (recommendation rule −perplexity_test_norm + npmi_mean + matched_cosine_mean, implemented in `data-pipeline/build_lda_sweep.py` of the companion code repository). This looks like a contradiction; it is not, and the resolution is that the two rules optimise different objectives.

F-4 recommended K. Maximises three intrinsic quality criteria: held-out perplexity, top-N word coherence, seed stability. On every labelled scene the F-4 grid shows perplexity monotonically increasing in K, topic-diversity monotonically decreasing in K, and matched-cosine essentially flat ($\in [0.95, 0.99]$); the rule therefore picks the smallest K in the grid, $K = 4$. This is the answer to *"what is the smallest basis that explains the corpus word frequencies well and is stable across seeds?"*

Canonical K. Answers a different question: *"what is the smallest basis that has any chance of being aligned with the $n_{classes}$ land-cover classes?"* A basis with $K < n_{\text{classes}}$ forecloses on the topic-routed-hard alignment: the model cannot predict more distinct classes than it has topics, so the per-class precision- recall table of Suppl F **??** would have at least $n_{\text{classes}} - K$ unreachable classes by construction. The canonical rule therefore lower-bounds K at n_{classes} (capped at 12 for parsimony) so that the downstream F-1, F-7, F-8 axes (classification, topic-label coupling, Hungarian alignment) are answerable on the same basis.

Implication for the framework. The two answers optimise competing notions of "best K" — F-4 maximises model parsimony, the canonical rule maximises downstream interpretability. The paper documents both rather than forcing a single choice. A second-paper redesign (Suppl G **??**) would lift the canonical K from a fixed parameter to a per-axis sweep ($K \in \{4, 6, 8, 10, 12, 16\}$ on every F-axis) to remove the asymmetry.

G.7 Axis F-5: paired ARI under masks

Detailed in conference supplement B; restated here for cross-reference. $\text{ARI}_{\text{paired}}(m) = \text{ARI}(\{z_d\}_\mathcal{S}, \{z_d^{(m)}\}_\mathcal{S})$.

H.8 Axis F-6: cross-method ARI

For each scene compute the partition set $\mathcal{P} = \{\text{label}, \text{topic-dom}, \text{Felz}, \text{SLIC-500}, \text{SLIC-2000}, \text{p-15}, \text{p-7}, \text{px}\}$. The reported matrix is

$$M_{ij} = \text{ARI}(\mathcal{P}_i, \mathcal{P}_j), \tag{16}$$

restricted to pixels with a valid (non-sentinel) value in both partitions. NMI matrix is computed analogously.

I.9 Axis F-7: topic–label coupling

For each (canonical topic t, label ℓ), $P(\ell \mid t) = N_{t,\ell}/\sum_{\ell'} N_{t,\ell'}$ with $N_{t,\ell}$ the count of pixels with dominant topic t and ground truth ℓ. Per-topic Shannon entropy: $H(L \mid t) = -\sum_\ell P(\ell \mid t) \log P(\ell \mid t)$. Per-topic KL divergence from the marginal: $\mathrm{KL}(P(L \mid t) \,\|\, P(L)) = \sum_\ell P(\ell \mid t) \log[P(\ell \mid t)/P(\ell)]$.

J.10 Axis F-8: per-topic Hungarian alignment

Confusion matrix $C \in \mathbb{Z}_{\geq 0}^{K \times K^{(m)}}$, $C_{k,j} = |\{d \in \mathcal{S} : z_d = k \wedge z_d^{(m)} = j\}|$. Hungarian: $\sigma^* = \arg\max_\sigma \sum_k C_{k,\sigma(k)}$, solved by `scipy.optimize.linear_sum_assignment` on $-C$. Swap rate: swap $= |\mathcal{S}|^{-1} \sum_d \mathbf{1}[z_d^{(m)} \neq \sigma^*(z_d)]$.

K.11 Axis F-9: HIDSAG preprocessing stability

For each HIDSAG subset, $R = 3$ preprocessing recipes $\{\text{cont-rem}, \text{deriv}, \text{smooth}\}$. For each recipe pair (r_a, r_b) compute $\mathrm{ARI}(\text{dom-top}_{r_a}, \text{dom-top}_{r_b})$ on the documents present in both recipes. Per-subset aggregate is the mean over the $\binom{3}{2} = 3$ pairs.

L.12 Axis F-10: cross-scene transfer

Given scene s with canonical basis $\phi^{(s)}$, project the documents of scene t through one E-step

$$\gamma_{d,k}^{(t \leftarrow s)} = \alpha + \sum_n \lambda_{d,n,k}^{(t \leftarrow s)}, \tag{17}$$

with λ computed from $\phi^{(s)}$ frozen. The transfer report is $\mathrm{ARI}(\text{label}_t, \arg\max_k \theta^{(t \leftarrow s)})$.

M.13 Axis F-11: rate–distortion

Reconstruct $\hat{\mathbf{x}}_d = \sum_k \theta_{d,k} \phi_k^\top$ in the band-frequency-count space, then de-quantise to a continuous reflectance via the inverse of the canonical tokenisation $\hat{x}_{d,b} = c_{d,b}^{\text{recon}}/Q$. Distortion is MSE $= D^{-1} \sum_d \|\hat{\mathbf{x}}_d - \mathbf{x}_d\|^2$ at rate $\log_2 K$ bits.

N.14 Axis F-12: external baseline

For each labelled scene, collect three reference OA values from prior published work. Report $\Delta_s = \mathrm{OA}_s^{\text{topic-routed-soft}} - \max_{r \in \text{refs}(s)} \mathrm{OA}_s^{(r)}$ with the per-paper split caveat.

O.15 Inverse Hungarian (used in F-8 swap-rate computation)

The post-permutation confusion matrix $\tilde{C}$ used for the journal Figure 3(b) panel is

$$\tilde{C}_{k,j} = C_{k,\sigma^{*-1}(j)}, \tag{18}$$

which moves the matched diagonal to the main diagonal. The right panel of journal Figure 3 visualises $\tilde{C}$ directly.